The correlation matrix

How similar every sample is to every other, in one square, with the similar ones brought next to each other.

AT A GLANCE

Where	Quality control → Correlation
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and at least two samples that are not entirely empty.
Data source	The data as loaded, on the $\log_2$ scale. There is no selector: quality control describes what the instrument produced, before filtering or imputation.
Coefficient	Pearson, over the features both samples of a pair quantified.
Ordering	Hierarchical, by complete linkage on $1 - r$ — not the order of the annotation, unless a correlation could not be computed.
Table	None.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

Every other quality-control page looks at samples one at a time. This one looks at pairs, and it answers the question those pages cannot: which samples resemble which. A set of replicates that belong together forms a block on the diagonal; a sample that resembles the wrong condition, or nothing at all, breaks the block, and that is visible here long before it becomes an unexplained result.

The rows are not in the order of the annotation. They are reordered so that similar samples sit next to one another, which is what turns the figure from a table of numbers into a picture of the experiment's structure.

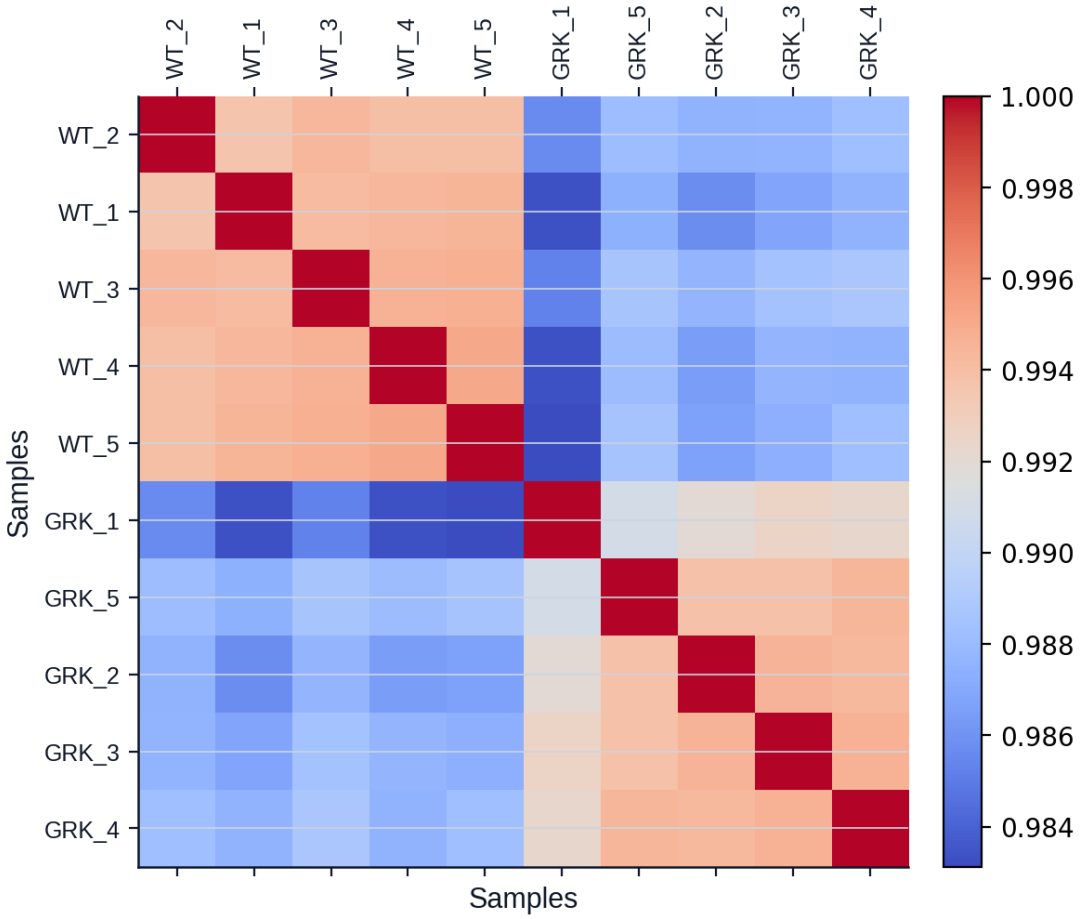


Figure 1.1. The ten samples of the example dataset at protein level, in Matrix mode. The clustering has put the five WT samples together and the five GRK samples together, so the two conditions appear as two blocks on the diagonal — and the colour bar, which runs from about 0.983 to 1, is the whole of the story: every pair correlates above 0.98, and the scale has been stretched across a range of two hundredths.

Reading it

- One cell, one ordered pair of samples — but the matrix is symmetric, so the cell above the diagonal and its mirror below carry the same number.
- The diagonal: every sample against itself, so exactly 1. It is the reference for the colour scale, not a result.
- The colour bar. Read its two ends before reading any cell: unless **Use Full Range** is on, they are the smallest and largest correlation actually observed, not −1 and 1.
- The order of the rows and columns, which is the dendrogram's, not the annotation's. Replicates that belong together end up adjacent; a sample that lands among another condition's replicates is the finding.

- A cell reading NA, in flat grey: that pair shares no feature that both quantified, so no correlation exists. When any such cell is present the clustering is skipped for the whole matrix and the samples stay in annotation order; the x-axis label says so.

EASY TO GET WRONG

With **Use Full Range** off — the default — the colour scale is stretched to the observed range, so a dataset in which every pair lies between 0.97 and 0.99 is painted from one end of the gradient to the other. That is useful for finding the least similar sample and disastrous for judging whether the samples are similar in absolute terms: the figure looks as dramatic at a range of 0.02 as at a range of 0.8. Turn **Use Full Range** on to fix the scale at −1 to 1 before concluding anything about the size of a difference. In the other direction, uniformly high correlations are not by themselves good news: proteomics samples correlate well because most of the proteome does not change, and a matrix with no structure at all can mean the conditions do not differ rather than that the data are clean.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to correlate: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Type	Matrix paints each cell with the colour of its correlation. Ellipse draws an ellipse per cell whose shape carries the magnitude and whose tilt carries the sign.	Matrix	page
Toggle IDs	Adds the original sample identifier in brackets after each label on both axes. It changes the labels only, never the ordering or the values.	off	page
Use Full Range	Fixes the colour scale at −1 to 1. With it off the scale spans the smallest and largest correlation in the matrix, which magnifies small differences.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. This plot starts taller and at a higher resolution than the others on the page, because the labels are small and numerous.	20 cm × 16 cm, 400	page

Controls on this page (continued)

Control	What it does	Default	Where
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Correlation gradient	Which colour scale the cells — or the ellipses — are painted from: Coolwarm, Red/blue, Spectral, Pink/green, Purple/green, Brown/teal, Viridis or Cividis. A diverging scale is the right choice while the range crosses zero; a sequential one is easier to read when every correlation is positive.	Coolwarm	appearance
Sample names	A display name per sample, for the figure only. Applied to the composite condition_n label, so a renamed condition follows through to these ticks.	as drawn	appearance
Title	This plot writes no title of its own. Entering one in the panel adds it, with the usual control over its size, weight and colour.	none	appearance
X axis, Y axis	Both labels read Samples and can be overwritten. The tick label size is set here too, which is the control that matters most on a wide matrix.	Samples, 7	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. This page offers no legend card and no condition colours: the colour bar is the legend, and the cells are not coloured by condition.

3 How it is computed

HOW IT IS COMPUTED

1. The matrix is the data as loaded, on the $\log_2$ scale. Non-numeric and non-finite entries are read as NA.
2. Samples are relabelled condition_n, numbered within the condition in the order the annotation gives, with the original identifier appended in brackets when **Toggle IDs** is on. A sample column that is entirely missing is dropped; at least two must remain.
3. For each pair of samples (a, b), the Pearson correlation over the features that both quantified:

$$r_{ab} = \frac{\sum_{i \in O_{ab}} (x_{ia} - \bar{x}_a)(x_{ib} - \bar{x}_b)}{\sqrt{\sum_{i \in O_{ab}} (x_{ia} - \bar{x}_a)^2} \sqrt{\sum_{i \in O_{ab}} (x_{ib} - \bar{x}_b)^2}},$$

   where O_{ab} is the set of features present in both. Pairs are handled independently, so different cells can rest on different numbers of features, and a pair sharing nothing gives NA.
4. The samples are reordered by hierarchical clustering: the distance between two samples is $d_{ab} = 1 - r_{ab}$, clipped to $[0, 2]$, the linkage is complete, and the rows and columns are permuted into the dendrogram's leaf order. If any correlation is NA the whole reordering is skipped and the annotation order is kept, because a distance matrix with a hole in it has no dendrogram.
5. The colour limits are the smallest and largest finite correlation, or -1 and 1 when **Use Full Range** is on. If every correlation is the same value, the fixed limits are used instead, since a scale of zero width paints nothing.

ELLIPSE MODE

The same matrix, drawn differently. Each cell holds an ellipse of fixed width whose height is $0.9\,(1 - |r|)$, so it collapses towards a line as $|r|$ approaches 1 and opens into a circle as r approaches 0. It is tilted $+45^\circ$ for a positive correlation and -45° for a negative one, and filled from the same colour scale as Matrix mode at 60 % opacity — so the shape and the colour say the same thing twice, which is what makes a small difference in a uniformly high matrix visible.

Look for samples whose relationships differ from comparable replicates, then inspect coverage and sample-level distributions. A high correlation does not establish agreement: adding a constant to every log intensity can leave Pearson's correlation unchanged. Shared differences between abundant and low-abundance proteins can also make every sample pair highly correlated. Distinct condition blocks are therefore neither required for a valid experiment nor proof of a treatment effect. Read the colour scale and the number of shared observations before comparing cells.

Edge cases

- Fewer than two non-empty sample columns leaves nothing to correlate, and the page reports that instead of drawing.
- A sample whose values are all identical has no variance, so every correlation with it is NA — one grey row and one grey column, and no clustering for the whole figure.
- Cells that are NA are painted flat grey and labelled NA in both modes, so an absent correlation is never mistaken for a low one.
- Pairwise handling means a cell can rest on very few shared features. A high correlation over two hundred features is not the same evidence as a high correlation over eight thousand, and the figure does not distinguish them; the missing-value heatmap does.
- The pairwise scatter of two conditions — the plot behind a correlation rather than its summary — is a separate page: Comparison → Pearson correlation.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = corr_plot(org_data, meta, id=False, data_scale="linear")
```

```r
fig <- corr_plot(org_data, meta, id = FALSE, data_scale = "linear")
```

data_scale declares the scale of the frame rather than converting it, and it decides what counts as missing, so the call above correlates linear intensities while the page correlates $\log_2$ ones — corr_plot(data, meta, id=False, data_scale="log2") is the call that reproduces the figure on this page. method="Ellipse" is **Type** Ellipse, and full_range=True is **Use Full Range**.

5 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. This is the one plot in the section where

the defaults are worth raising rather than lowering — with twenty or more samples the tick labels need the height. The Matrix cells are drawn as one vector shape each, so an SVG or PDF download contains no embedded image and each cell can be selected in a drawing program. Choose PDF for a figure that will be scaled or edited, PNG for a slide.

Behind this page

Function	What it produces
corr_plot	The correlation matrix, in either mode, with the hierarchical reordering